

IPCop

A Nostalgic Look



Part—6

Since December 2011, over a series of five articles in LFY, I have covered several important aspects of IPCop firewall version 1.4.21, including its inbuilt features and various add-ons that raise it to almost commercial levels. Now, though a new stable release of IPCop – 2.0 -- has arrived, let us continue our discussion on the 1.4.21 distro and explore the new IPCop release next month onwards.

Needless to say, IPCop is a great and absolutely stable stateful packet inspection (SPI) firewall distro. Being an SPI firewall, it keeps track of all outgoing requests and subsequently allows incoming traffic only as replies towards the corresponding outgoing request. This ensures no unauthorised communication can be initiated from the Internet towards the internal network.

The installation of IPCop is simple and straightforward once you understand its interfaces – which are coloured in an intuitive way. Red denotes the untrusted network, typically the Internet; green is the protected internal network; blue is typically used as a wireless subnet and orange is the demilitarised zone or DMZ. Default access in between various zones is predefined to allow only controlled traffic as detailed in Table 1.

The default scheme blocks all access from the Internet (red), the DMZ (orange) and wirelessly (blue) towards the protected network (green). Further, the VPN can be used to allow secure access from blue to green.

Apart from blocking unintended traffic, a good firewall must also have several other functions such as VPN, logging, a time server, etc. IPCop does support various such functions;

a list is given in Table 2.

Though an excellent firewall distro, IPCop lags behind on several important functions such as categorised URL filtering, OpenVPN, proxy reports, etc. Please refer to Table 3 for details of several add-ons, which integrate various functions in IPCop. Please refer to the earlier articles on IPCop in LFY for detailed instructions on installation, configuration and troubleshooting.

Access to IPCop is available on non-standard TCP ports – SSH on 222 and https on 445. To transfer binaries of various add-ons to the IPCop box, those on Linux systems can use default SSH and SCP commands, whereas those on Windows systems can use free downloadable tools, putty and WinSCP, respectively.

Apart from these, the various other add-ons available for IPCop include:

- Copfilter – A gateway antivirus
 - Ntop – For network monitoring
 - p2pblook – Blocks various peer-to-peer downloads
 - WLAN AP – Converts IPCop to a wireless access point
- IPCop 1.4.21 has helped various organisations to secure their network since it was released on July 24, 2008. Since

Table 1: The IPCop access matrix

Description	Source interface	Destination interface	Default Access
Internet/untrusted	Red	Green, blue	Blocked
		Orange	Allowed
Hosts accessible from the Internet, semi-trusted	Orange	Red	Allowed
		Green and blue	Blocked *
Wireless subnet	Blue	Green	Blocked
		Red/orange	Based on the MAC address and/or IP address
Protected	Green	Red/orange	Allowed
		Blue	Allowed **

* May be allowed by configuring the DMZ pinholes

** May require configuration of blue to green pinholes